

UEL Property ABI Card Audit: f42 Parameter Mapping Order vs Solve Decks  
(Explaining 28,000x Modulus Inversion in Job 1409947)

| Parameter Index | f42_mixed_uel.for Variable      | Intended Value (Physical) | Job 1409734 (Ref 15k Decks) | Job 1409947 (Adapt 14k Decks) | Status / Ratio           |
|-----------------|---------------------------------|---------------------------|-----------------------------|-------------------------------|--------------------------|
| PROPS(1)        | E_L0 (Length scale l0)          | 0.0075 mm                 | 0.0075                      | 210.0 mm                      | 28,000x larger (diffuse) |
| PROPS(2)        | E_GC (Fracture energy Gc)       | 0.0027 kN/mm              | 0.0027                      | 0.3 kN/mm                     | 111x tougher             |
| PROPS(3)        | E_MOD (Young's modulus)         | 210.0 kN/mm2              | 210.0                       | 0.0075 kN/mm2                 | 28,000x softer           |
| PROPS(4)        | E_NU (Poisson's ratio nu)       | 0.30                      | 0.30                        | 0.0027                        | 111x lower               |
| PROPS(5)        | E_K (Residual stiffness k)      | 1.0e-7                    | 1.0e-7                      | 1.0e-7                        | Matched                  |
| PROPS(6)        | N_PHYS (Physical element count) | 14,483 / 15,192           | 15,192.0                    | 1.0                           | Unscaled / Diagnostic    |

Discovery: Job 1409947 used Molnar order (E, nu, l0, Gc) while f42 subroutine expects (l0, Gc, E, nu, k, N\_phys).  
Job is running safely to full termination per protocol before qualified resubmission.